

THE FLORIDA STATE UNIVERSITY

COLLEGE OF ENGINEERING

**BIOFABRICATION OF RHEOLOGICALLY TUNED IN VITRO TISSUE
MODELS FOR EVALUATING EARLY-STAGE MUSCLE REPAIR**

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2. ABSTRACT

Skeletal muscle injuries are among the most prevalent musculoskeletal conditions worldwide, yet current *in vitro* models lack the physiological fidelity needed to capture the dynamic interplay between mechanical and immune-derived cues that govern early-stage tissue repair. Existing two-dimensional cell culture systems and conventional hydrogel platforms fail to simultaneously recapitulate the stiffness of the early regenerative extracellular matrix and the paracrine signaling environment established by infiltrating macrophages. This thesis addresses that gap by developing and characterizing a rheologically tuned alginate hydrogel platform integrated with macrophage-conditioned media (MCM) to model the early muscle regeneration niche *in vitro*.

C2C12 murine myoblasts were first characterized in two-dimensional culture under defined serum conditions, 10% fetal bovine serum (FBS) for proliferation and 0% and 2% FBS for differentiation, to establish a reproducible baseline for myogenic progression. The influence of macrophage-conditioned media on differentiation was subsequently evaluated, revealing that while low-serum conditions reliably promoted myotube formation, MCM supplementation disrupted myogenic progression in 2D culture, underscoring the context-dependence of immune-derived signals. These findings motivated the transition to a three-dimensional system in which biophysical cues could be precisely controlled.

Alginate hydrogels were fabricated using low-molecular-weight alginate crosslinked with calcium chloride and characterized by oscillatory rheology. Stiffness was tuned by varying polymer concentration and molecular weight to achieve storage moduli spanning approximately 2 kPa, encompassing the physiologically relevant stiffness range of early post-injury muscle ECM. C2C12 myoblasts encapsulated within these 3D constructs maintained high viability over seven days of culture even in the presence of MCM, demonstrating that the alginate microenvironment

provides a cytoprotective niche that buffers cells against the disruptive effects of immune-derived factors observed in 2D.

Finally, extrusion-based 3D bioprinting parameters, including extrusion rate, nozzle speed, and crosslinking conditions, were systematically optimized to achieve structurally consistent, scalable alginate filaments. Printability was quantified using a printability index comparing deposited filament width to the optimal filament width, identifying conditions that yield high-fidelity constructs suitable for tissue engineering applications.

Collectively, this work establishes a biofabrication-compatible, mechanically tunable alginate platform that integrates biochemical and biophysical cues relevant to early skeletal muscle repair. The resulting model provides a reproducible and scalable framework for studying immune–myoblast communication and for evaluating candidate regenerative therapies without reliance on animal models.

3. INTRODUCTION AND LITERATURE REVIEW

3.1 Clinical and Biological Significance of Skeletal Muscle Injury

Skeletal muscle constitutes approximately 40% of total body mass and is essential for locomotion, posture, metabolic regulation, and thermoregulation [1]. Despite its remarkable intrinsic capacity for regeneration, skeletal muscle remains one of the most commonly injured tissues in both athletic and clinical populations. Muscle injuries account for approximately 35–55% of all sports-related injuries, with recurrence rates reported as high as 30% following return to activity [2]. These high recurrence rates are widely attributed to incomplete tissue healing, premature return to loading, and a poor understanding of the underlying repair dynamics at the cellular and matrix levels. Beyond sports medicine, skeletal muscle injuries and degenerative

myopathies, including Duchenne muscular dystrophy, sarcopenia, and volumetric muscle loss (VML) resulting from traumatic injury or surgical resection, affect millions of individuals globally and impose significant burdens on healthcare systems [3]. Despite extensive clinical interest, current therapeutic strategies for muscle repair remain largely conservative and non-specific, relying primarily on rest, physical therapy, and in severe cases, surgical intervention. The development of more targeted regenerative strategies, including growth factor delivery, cell-based therapies, and engineered tissue constructs, requires a thorough mechanistic understanding of early muscle repair. However, this understanding is constrained by the limited physiological relevance of current experimental models. *In vivo* animal models, while informative, are costly, subject to interspecies variability, and difficult to control for individual variables such as stiffness or cytokine concentration [4]. This has motivated the development of *in vitro* tissue engineering platforms capable of recreating the complexity of the muscle repair environment in a reproducible and controllable manner.

3.2 The Biology of Skeletal Muscle Repair: From Injury to Regeneration

Skeletal muscle regeneration following injury is a highly orchestrated, multiphasic process that depends on the coordinated interaction of immune cells, resident muscle stem cells (satellite cells), and the extracellular matrix (ECM) [5]. Understanding each phase of this process is essential for designing biomimetic *in vitro* models that accurately recapitulate the early repair environment. The first phase, occurring within hours of injury, is characterized by disruption of the sarcolemma, release of intracellular contents, and rapid infiltration of innate immune cells [5]. Neutrophils are among the first responders, followed by classically activated pro-inflammatory macrophages (M1 phenotype). These M1 macrophages, which predominate during the first 24-72 hours post-injury, release a suite of pro-inflammatory cytokines including tumor necrosis factor-alpha (TNF- α),

interleukin-1 β (IL-1 β), and interleukin-6 (IL-6) [6]. Critically, these signals serve not only to clear cellular debris and necrotic tissue through phagocytosis but also to stimulate satellite cell activation and myoblast proliferation. This pro-inflammatory phase is therefore not merely destructive. It is a necessary prerequisite for effective regeneration.

Between approximately 48 hours and one week post-injury, the inflammatory milieu shifts toward an anti-inflammatory, pro-regenerative state. Macrophages undergo phenotypic transition, often referred to as M2 polarization, driven by signals including interleukin-4 (IL-4), interleukin-13 (IL-13), and interleukin-10 (IL-10) [6]. M2 macrophages secrete growth factors such as insulin-like growth factor-1 (IGF-1), transforming growth factor-beta (TGF- β), and vascular endothelial growth factor (VEGF), which collectively promote myoblast differentiation, myotube fusion, and ECM remodeling [7]. The sequential M1-to-M2 transition is now recognized as a master regulator of muscle regeneration; disruption of this transition, for example, by premature suppression of inflammation or failure to resolve the pro-inflammatory state, results in incomplete regeneration, fibrosis, and impaired muscle function [6].

Concurrent with the immune response, quiescent satellite cells, the primary myogenic stem cell population in adult skeletal muscle, become activated in response to both mechanical damage and paracrine cues from inflammatory cells [5]. Once activated, satellite cells re-enter the cell cycle, undergo extensive proliferation, and eventually commit to terminal differentiation as myoblasts. Myogenic differentiation is regulated by a family of transcription factors known as myogenic regulatory factors (MRFs), including MyoD, myogenin, Myf5, and MRF4 [8]. Upregulation of myogenin and MyHC marks the transition from proliferating myoblasts to post-mitotic myocytes, which subsequently fuse to form multinucleated myotubes. These myotubes

undergo progressive maturation, forming organized sarcomeres and eventually reconnecting with the surrounding ECM to restore functional muscle architecture [5].

The ECM itself plays an indispensable role throughout this process. Native skeletal muscle ECM is a hierarchically organized structure composed primarily of collagen types I, III, and IV, laminin, fibronectin, and proteoglycans, with a shear modulus in the range of approximately 2–12 kPa depending on the degree of contraction and the specific muscle studied [9]. Following injury, the mechanical properties of the ECM change dynamically: the immediate post-injury microenvironment is softer (~2–4 kPa) due to matrix degradation and edema, while later-stage fibrosis can result in stiffening beyond the physiological range [9]. These mechanical signals are transduced by myoblasts through integrin-mediated focal adhesion complexes and downstream mechanosignaling pathways including YAP/TAZ and MRTF-A, directly influencing myogenic commitment, alignment, and fusogenic capacity [10]. Consequently, any physiologically relevant *in vitro* model of early muscle repair must account for both the biochemical (immune-derived) and biophysical (matrix stiffness) dimensions of the regenerative niche.

3.3 Current In Vitro Models: Capabilities and Gaps

The field of skeletal muscle tissue engineering has produced a range of model systems with varying degrees of biological fidelity. The simplest and most widely adopted approach is two-dimensional (2D) monolayer culture of myogenic cell lines such as C2C12 murine myoblasts on rigid tissue culture plastic or ECM-coated surfaces. In this system, differentiation is reliably induced by reducing the concentration of fetal bovine serum (FBS) from proliferative levels (10% FBS) to low-serum differentiation media (1–2% FBS), which drives cells to exit the cell cycle and activate MRF-mediated differentiation programs [11]. While 2D systems have provided foundational insights into myogenic differentiation and the role of individual signaling molecules,

they fail to recapitulate critical features of the native muscle microenvironment: cells grow on surfaces that are orders of magnitude stiffer than native muscle tissue, lack a three-dimensional architectural context, and are not subjected to physiologically relevant mechanical or immune cues in an integrated manner [4].

To address the limitations of 2D culture, researchers have developed hydrogel-based 3D scaffolds with tunable mechanical properties. Collagen and fibrin gels were among the first materials explored; while they offer inherent cell-adhesion ligands and support myotube formation, they suffer from limited mechanical tunability, rapid enzymatic degradation, and significant batch-to-batch variability [12]. Polyethylene glycol (PEG) hydrogels offer precise control over crosslink density and stiffness, but their bioinert nature necessitates functionalization with adhesion peptides such as RGD to support cell attachment, adding synthetic complexity [12]. Gelatin methacryloyl (GelMA) hydrogels have emerged as particularly promising due to their photocrosslinkable nature and the retention of cell-binding domains from native gelatin; studies using GelMA have demonstrated that myoblasts exhibit improved alignment and fusion on matrices with elastic moduli in the 1–10 kPa range [10]. However, UV crosslinking introduces potential phototoxicity concerns, and the photopolymerization process can be difficult to standardize across laboratories.

Alginate-based hydrogels represent an attractive alternative for engineering physiologically relevant muscle constructs. Derived from brown seaweed, alginate is a naturally occurring polysaccharide that undergoes rapid ionic gelation in the presence of divalent cations such as calcium (Ca^{2+}), making it highly suitable for cell-compatible crosslinking without chemical initiators or UV exposure [13]. The stiffness of alginate hydrogels can be tuned across a broad range by varying the polymer concentration, the molecular weight of the alginate, and the

crosslinker concentration, enabling the fabrication of matrices spanning from soft (~0.5 kPa) to relatively stiff (~50 kPa) formulations [13]. Importantly, alginate is also compatible with extrusion-based 3D bioprinting, making it a promising bioink material for scalable fabrication of complex tissue constructs [14].

The integration of immune cell-derived cues into engineered muscle models represents a critical and underexplored frontier. While macrophage-conditioned media (MCM) has been used in a number of studies to approximate the paracrine environment of the regenerating muscle niche, the majority of hydrogel-based models have focused exclusively on mechanical cues, without incorporating the dynamic immune signaling that is essential for authentic recapitulation of early repair [7]. A small number of co-culture studies have demonstrated that macrophage-secreted factors can significantly enhance myotube formation and maturation *in vitro*, particularly those derived from M2-polarized macrophages [15]. Yet the combination of rheologically tuned hydrogels with macrophage-conditioned media in a unified 3D model system has not been systematically investigated. This represents a significant gap in the field, given that in the native regenerative niche, biochemical and biomechanical signals act simultaneously and synergistically.

Table 1 below summarizes the landscape of current *in vitro* model systems for skeletal muscle tissue engineering, comparing their materials, mechanical properties, capacity for immune integration, and key limitations.

Model Type	Key Materials	Mechanical Range	Immune Integration	Key Limitations
2D Monolayer Culture	Tissue culture plastic, ECM coatings (laminin, fibronectin)	Rigid (~GPa); not tunable	Conditioned media (indirect)	No 3D architecture; does not replicate ECM stiffness or cell embedding
Fibrin/Collagen Gels	Fibrin, collagen type I, Matrigel	0.1–2 kPa; limited upper range	Some co-culture studies reported	Batch variability; rapid degradation; low printability
GelMA Hydrogels	Gelatin methacrylate, UV crosslinking	1–100 kPa (tunable by %)	Rare; mostly mechanical focus	Requires UV crosslinking; potential phototoxicity to cells
PEG Hydrogels	Polyethylene glycol, functionalized crosslinkers	0.1–100 kPa (highly tunable)	Limited; bioinert unless functionalized	Bioinert without modification; complex synthesis; no inherent cell-binding ligands
Alginate Hydrogels	Alginate (low/high MW), CaCl ₂ ionic crosslinking	0.5–50 kPa (tunable by concentration and MW)	Largely unexplored; this study integrates MCM	Limited cell-adhesion motifs; diffusion constraints in thick constructs
Bioprinted 3D Constructs (GelMA/alginate)	Bioprinted hydrogels, electrospun scaffolds, decellularized ECM	Variable; dependent on bioink formulation	Rarely combined with immune signals	Complex fabrication; printing parameters require optimization; immune cues rarely integrated

Table 1: Comparison of common *in vitro* model systems used for skeletal muscle tissue engineering, highlighting mechanical range, immune integration capacity, and key limitations.

3.4 Rational for Present Study

Taken together, the existing literature establishes three key variables that must be integrated to produce a physiologically relevant *in vitro* model of early muscle repair: (i) serum concentration as a defined biochemical stimulus for myogenic differentiation; (ii) macrophage-derived conditioned media as a proxy for the dynamic immune microenvironment; and (iii) hydrogel stiffness as a tunable biophysical cue that modulates myoblast mechanotransduction and matrix remodeling. Individually, each of these variables has been studied to varying degrees.

However, no study to date has systematically combined all three within a single, bioprinting-compatible platform.

The present thesis addresses this gap by developing and characterizing a rheologically tuned alginate hydrogel system seeded with C2C12 myoblasts and supplemented with macrophage-conditioned media, with the ultimate goal of fabricating a 3D bioprinted construct that recapitulates the early muscle regeneration niche. By integrating biochemical and biophysical cues within a scalable and reproducible framework, this work aims to establish a next-generation platform for investigating immune–muscle communication and for screening regenerative therapeutics without reliance on animal models.

4. THESIS STATEMENT

Skeletal muscle possesses a remarkable intrinsic capacity for self-repair; however, this regenerative potential is not unlimited and is critically dependent on the coordinated sequence of cellular and molecular events that define the early post-injury microenvironment. The transition from a pro-inflammatory to a pro-regenerative state, orchestrated largely by macrophage phenotypic switching, must occur within a precisely defined mechanical context shaped by the extracellular matrix. When either the biochemical or biophysical dimensions of this niche are disrupted, regeneration is incomplete, and functional recovery is compromised. Despite decades of research into skeletal muscle biology, the tools available to study this process *in vitro* remain inadequate. Conventional two-dimensional monolayer systems cannot replicate the three-dimensional architecture or matrix stiffness of native muscle, while most hydrogel-based models address mechanical cues in isolation, without incorporating the immune-derived paracrine signals that are equally essential to the repair process. The result is a persistent gap between our

mechanistic understanding of muscle regeneration and our ability to model it faithfully in a controlled laboratory setting.

This thesis takes the position that bridging this gap requires the simultaneous integration of two classes of microenvironmental cues: biophysical cues, in the form of a mechanically tunable scaffold, and biochemical cues, in the form of macrophage-derived signaling. Alginate hydrogels, crosslinked ionically with calcium chloride, offer a compelling platform for this integration. Their stiffness can be tuned across the physiologically relevant range of early post-injury muscle ECM by modulating polymer concentration and molecular weight, and their compatibility with extrusion-based 3D bioprinting makes them scalable for tissue fabrication. When supplemented with macrophage-conditioned media, these hydrogels can approximate the paracrine landscape of the regenerating muscle niche, providing a multifactorial *in vitro* environment that no prior model has achieved within a single, print-ready platform. Critically, the 3D context of the hydrogel changes how myoblasts respond to immune-derived signals compared to 2D culture, revealing that the interplay between matrix stiffness and inflammatory cues is more nuanced than either factor alone would suggest.

The experimental work presented in this thesis was designed to test whether this integrated approach is both feasible and scientifically productive. Beginning with a rigorous characterization of myogenic differentiation under defined serum conditions, and progressing through macrophage-conditioned media studies, rheological optimization of alginate formulations, 3D encapsulation of C2C12 myoblasts, and extrusion-based bioprinting, each phase builds toward a unified and reproducible platform. The results demonstrate that stiffness-tuned alginate hydrogels preserve myoblast viability and create a cytoprotective 3D niche even in the presence of macrophage-derived factors that disrupt differentiation in 2D—suggesting that

matrix context is a critical variable in how muscle precursor cells interpret immune signals. Together, these findings support the following central thesis: A rheologically tuned alginate hydrogel combined with macrophage-conditioned media can better model early-stage muscle repair by promoting myogenic differentiation *in vitro*.

5. MATERIALS AND METHODS

5.1 2D Cell Culture

5.1.1 Cell Lines and Maintenance

C2C12 murine myoblasts (ATCC CRL-1772) were maintained in standard proliferation medium consisting of Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and 1% penicillin–streptomycin. Cells were cultured in T-75 flasks at 37°C with 5% CO₂, with media changes every two to three days. All experiments were initiated when cultures reached approximately 80–90% confluency to ensure consistent differentiation induction across conditions.

RAW 264.7 murine macrophages were cultured in DMEM supplemented with 10% FBS and 1% penicillin–streptomycin under identical incubation conditions. After 24 hours of culture, the media was changed to 2% FBS to match the differentiation media for C2C12 myoblasts. After 48 hours, macrophage-conditioned media (MCM) was collected and to use in downstream differentiation experiments.

5.1.2 Assessment of FBS Concentration on Myoblast Differentiation

To identify the optimal serum concentration for driving myogenic differentiation, C2C12 myoblasts were first expanded in DMEM containing 10% FBS. Upon reaching 80–90% confluency, cells were trypsinized, counted, and replated in 6-well plates at a standardized seeding density. On Day 0, growth media was replaced with differentiation media containing 0%, 2%, or 10% FBS to assess the effect of progressive serum reduction on myogenic commitment and myotube formation. The 10% FBS condition served as a non-differentiating proliferative control. Differentiation media was replenished on Day 6, and image acquisition was performed on Day 7 using the EVOS M5000 fluorescence microscope at 10× and 20× magnification.

5.1.3 Assessment of Macrophage-Conditioned Media on Myoblast Differentiation

To evaluate the paracrine influence of macrophage-secreted factors on myogenic differentiation, C2C12 myoblasts were cultured in media supplemented with varying concentrations of RAW 264.7 macrophage-conditioned media (MCM). Following collection and filtration, MCM was combined with the optimal serum-starvation condition identified in Section 5.1.2 to produce differentiation media containing 30%, 50%, or 100% MCM. A control condition without MCM was included in parallel. Cells were plated on Day 0 and treated with MCM-supplemented differentiation media from Days 1 through 6. On Day 7, live-cell fluorescence images were acquired using the Incucyte® Live-Cell Analysis System (Sartorius), which enabled automated, high-content imaging of cell confluence and morphology in a non-invasive manner across all MCM conditions throughout the differentiation window.

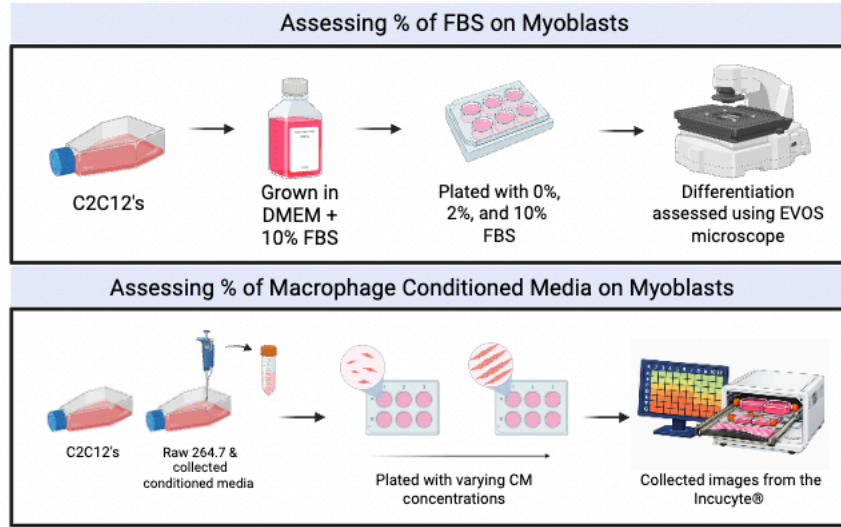


Figure 1: Schematic overview of 2D myoblast culture experimental workflow.

5.2 Data Analysis

Myotube quantification from phase-contrast images was performed using TrueFad, a deep learning-based image analysis pipeline implemented through the DeepImageJ plugin in Fiji/ImageJ (Proux-Gillardeaux et al., 2023). The workflow proceeded in four sequential steps. First, input phase-contrast images were scaled to a standardized resolution of 512×512 pixels to ensure compatibility with the trained model. Second, the rescaled images were passed through the DeepImageJ deep learning model, which generated a probability map highlighting myotube-like elongated structures. Third, the prediction output underwent a post-processing pipeline consisting of pixel intensity thresholding to binarize the prediction map, followed by morphological post-processing and watershed segmentation to separate adjacent and overlapping myotubes into individually labeled objects. Fourth, edge-contacting objects were trimmed from the field of view to avoid partial measurement errors, and regions of interest (ROIs) were

generated for each detected myotube. Morphological measurements including myotube count, length, area, and orientation were exported to Microsoft Excel for statistical analysis. This pipeline provided an objective, automated approach to quantifying myogenic differentiation outcomes across all experimental conditions.

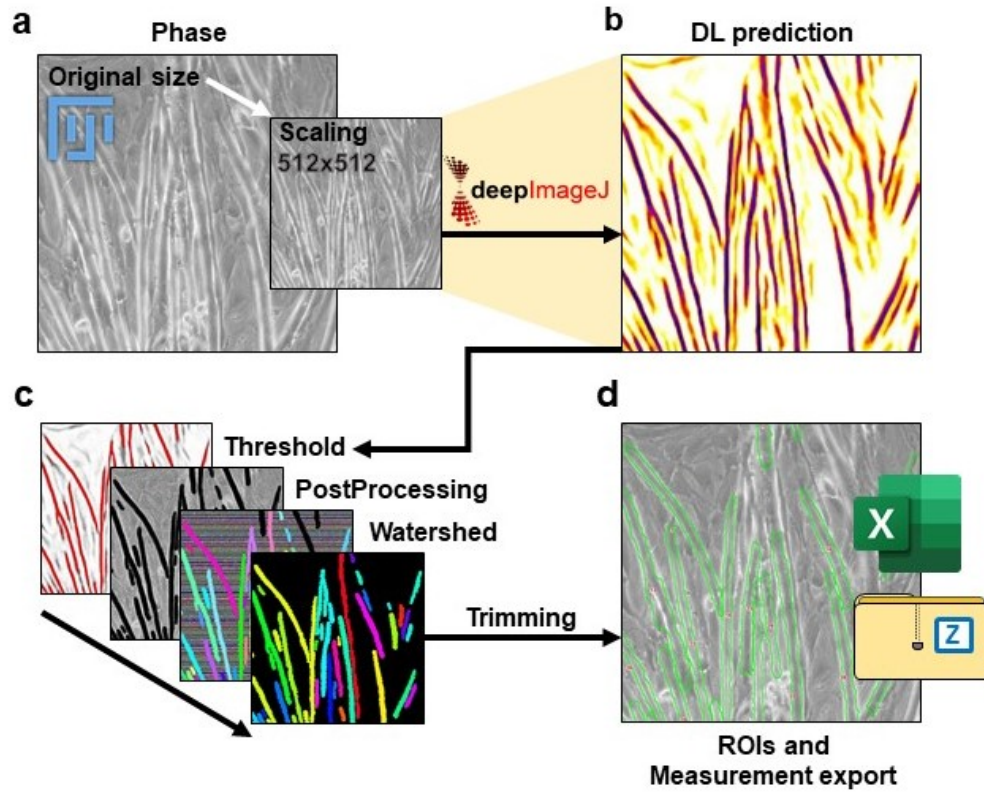


Figure 2: Deep learning-based image analysis pipeline for myotube quantification.

5.3 Hydrogels

5.3.1 Hydrogel Fabrication

Alginate hydrogels were prepared using low-viscosity alginate (Thermo Fisher Scientific) at a concentration of 0.8% w/v as the base formulation. Alginate powder was dissolved in serum-free DMEM and homogenized using a THINKY centrifugal mixer (ARE-310) at 2000 rpm for

30 minutes to ensure uniform polymer distribution and minimize air inclusion. Ionic crosslinking was achieved by exposing the hydrogel to a 5% calcium chloride (CaCl_2) solution, which initiates rapid gelation through divalent cation exchange with the alginate polymer chains. This crosslinking method was selected for its cell compatibility, as it requires neither UV exposure nor chemical initiators.

5.3.2 Rheological Characterization

The viscoelastic mechanical properties of the fabricated hydrogels were characterized using oscillatory rheology. Frequency sweeps were performed to measure the storage modulus (G') and loss modulus (G'') as a function of angular frequency. A stable gel is indicated when G' exceeds G'' across the tested frequency range, confirming solid-like elastic behavior. Hydrogel formulations were evaluated to identify compositions with storage moduli within the physiologically relevant range of 2–10 kPa, approximating the stiffness of early post-injury skeletal muscle extracellular matrix. Results were used to guide iterative optimization of alginate concentration and molecular weight.

5.3.3 Printability Assessment

To evaluate the suitability of the hydrogel formulation as a bioink for extrusion-based 3D bioprinting, hydrogels were printed across a matrix of nozzle speeds and extrusion rates using a benchtop bioprinter. Printed filaments were imaged using brightfield microscopy and analyzed for line width consistency and structural fidelity. Printability was quantified using a Printability Index, defined as:

$$\text{Printability Index} = \frac{\text{Line Width}}{\text{Optimal Width}}$$

The optimal width was determined as the width that produced prints with the most structural integrity. A Printability Index approaching 1.0 indicates high-fidelity extrusion with deposited filament width matching the theoretical nozzle diameter. Conditions yielding a Printability Index closest to 1.0 were identified as optimal printing parameters for downstream cell-laden construct fabrication.

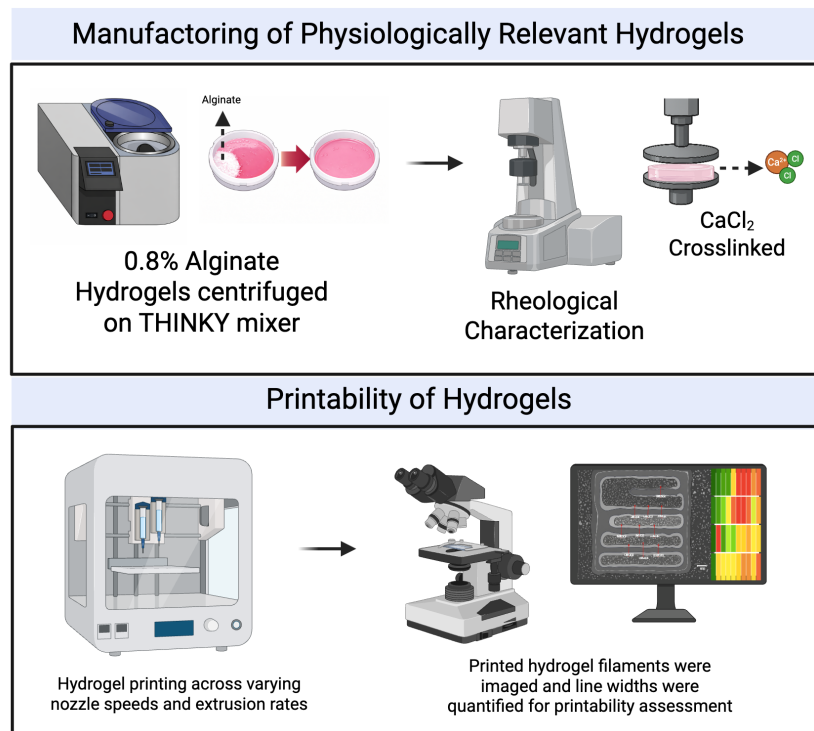


Figure 3: *Fabrication and characterization of alginate-based hydrogels.*

5.4 3D Cell Culture

To evaluate whether macrophage-conditioned media provides a supportive environment for myoblast attachment and survival within a 3D context, C2C12 myoblasts were encapsulated in alginate hydrogels formulated with MCM. Hydrogels were prepared as described in Section 5.3.1 and seeded with 3×10^6 C2C12 myoblasts per construct. Two experimental groups were included: a control condition using hydrogels formulated with 2% FBS differentiation media only, and an MCM condition using hydrogels formulated with 50% MCM. Culture media was supplemented throughout the 7-day differentiation window with matching conditioned media concentrations to maintain consistent biochemical exposure. Following 7 days of culture, cell viability was assessed using a Live/Dead fluorescence assay. Cultures were incubated with Calcein-AM (live cells, green) and Ethidium Homodimer-1 (dead cells, red) for 30 minutes at 37°C, and fluorescence images were acquired using the EVOS M5000 imaging system. Live fraction was quantified from fluorescence images as the ratio of live-stained cells to the total cell population per field of view.

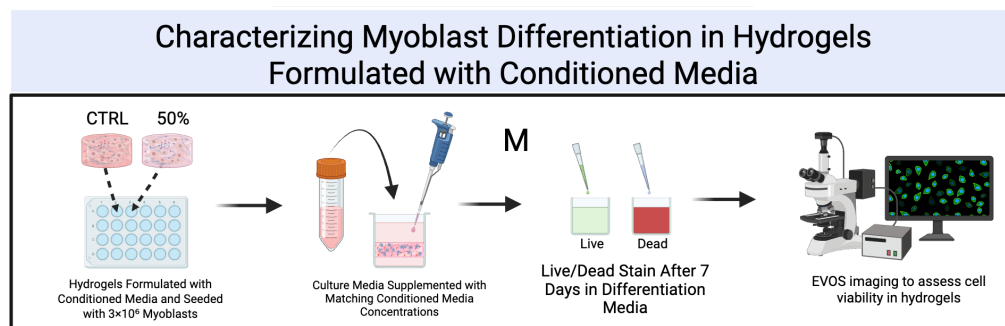


Figure 4: *Experimental workflow for characterizing myoblast differentiation within conditioned media-supplemented hydrogels.*

6. RESULTS AND DISCUSSION

The results of this study are organized to reflect the sequential experimental logic underlying the development of a rheologically tuned in vitro skeletal muscle model. Beginning with two-dimensional culture, the effects of serum concentration and macrophage-conditioned media on C2C12 myoblast differentiation were systematically characterized to establish a reproducible biochemical baseline. These findings motivated the transition to three-dimensional alginate hydrogel constructs, where stiffness was tuned across the physiologically relevant range of early post-injury muscle ECM and cell viability was assessed in the presence of conditioned media. Finally, extrusion-based bioprinting parameters were optimized and evaluated through quantitative printability analysis. Together, these results address the central hypothesis that integrating immune-derived biochemical cues with mechanically tuned biofabricated constructs can recapitulate key features of the early skeletal muscle regeneration niche in vitro.

6.1 Low Serum Promotes Differentiation, While Conditioned Media Disrupts Myogenic Progression in 2D Culture

Two-dimensional culture experiments were conducted in two sequential phases to characterize the biochemical conditions governing C2C12 myoblast differentiation and to evaluate how macrophage-derived paracrine signals modulate that process. In the first phase, C2C12 myoblasts were cultured under three serum concentrations — 0%, 2%, and 10% FBS — to determine the optimal conditions for inducing myogenic progression while maintaining cell viability. In the second phase, the effect of macrophage-conditioned media (MCM) collected from RAW 264.7 macrophages was assessed at concentrations of 30%, 50%, and 100% in C2C12 cultures maintained under the differentiation condition identified in phase one. Cell morphology was monitored longitudinally using the Incucyte® Live-Cell Analysis System.

Myotube formation was quantified by applying a watershed-based segmentation pipeline in TrueFad and DeepImageJ to calculate fusion index across experimental conditions.

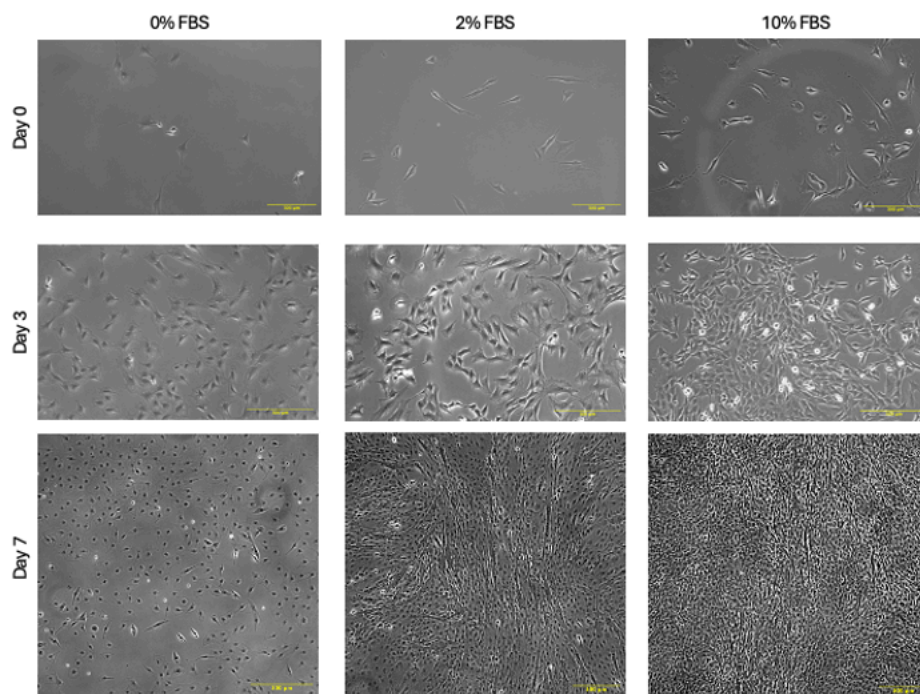


Figure 5: *C2C12 morphology over time under varying serum conditions.*

Phase-contrast imaging was used to longitudinally assess C2C12 myoblast morphology and confluence across three serum conditions — 0%, 2%, and 10% FBS — at Days 0, 3, and 7 of culture (Figure 5). At Day 0, all three conditions displayed comparably sparse populations of spindle-shaped myoblasts with no observable differences in morphology, confirming a consistent seeding baseline prior to differentiation induction. By Day 3, condition-dependent divergence began to emerge. Cultures maintained in 0% FBS showed minimal proliferation and remained low-density, consistent with the absence of serum-derived mitogenic signals, though individual cells retained an elongated morphology. The 2% FBS group displayed increased confluence relative to 0% FBS, with cells beginning to align and establish the cell-cell contacts necessary to initiate fusion. The 10% FBS condition showed the greatest proliferative expansion, producing

dense monolayers characteristic of a growth-promoting environment. By Day 7, these distinctions were markedly pronounced. The 0% FBS cultures remained sparse with scattered, poorly aligned cells and visible evidence of reduced viability, indicating that complete serum withdrawal is too cytotoxic to support productive differentiation. The 2% FBS group, by contrast, formed a confluent cell sheet with clearly visible elongation and parallel alignment, morphological hallmarks consistent with myogenic progression and early myotube organization. The 10% FBS cultures reached hyper-confluence and, while some aligned cell clusters were apparent, the overall phenotype was dominated by densely packed proliferating cells rather than the organized elongated structures associated with terminal differentiation. Taken together, these results demonstrate that 2% FBS provides the optimal balance of survival and differentiation-permissive signaling among the conditions tested, and established it as the baseline serum condition for all subsequent experiments.

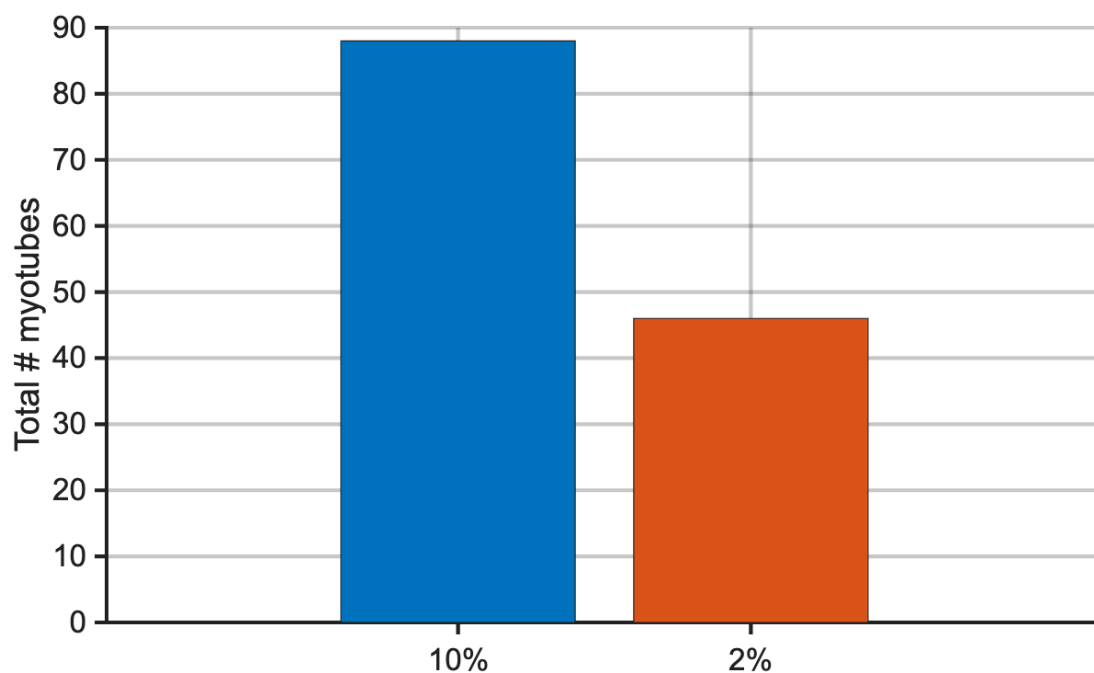


Figure 6: *Quantification of total myotube count at Day 7 for 2% and 10% FBS conditions across two fields of view*

To quantify myogenic output beyond morphological observation, myotube formation was assessed at Day 7 using a watershed-based segmentation pipeline in TrueFad and DeepImageJ applied to phase-contrast images collected from 2% and 10% FBS cultures. Total myotube count across two fields of view revealed a higher number of myotubes in the 10% FBS condition compared to 2% FBS (Figure 6). However, this difference is best interpreted in the context of cell density rather than differentiation efficiency. As established by the morphological data (Figure 5), 10% FBS cultures reached substantially greater confluence by Day 7 due to sustained mitogenic signaling, meaning that a larger absolute pool of cells was available for fusion events. The 2% FBS condition, while yielding a lower total myotube count, supported this process from a more controlled starting density where serum withdrawal had already shifted cells toward a differentiation-competent state. Myotube count alone, therefore, does not capture the relative efficiency of differentiation and must be considered alongside the morphological and quantitative metrics described below.

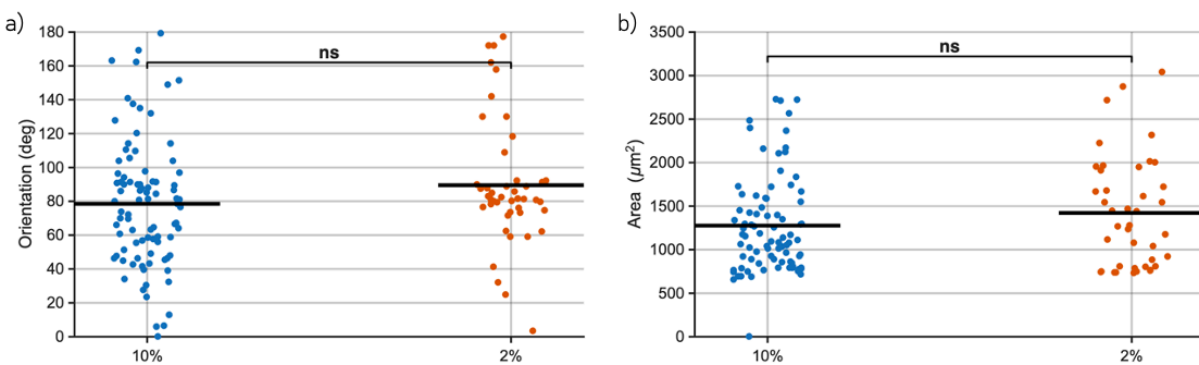


Figure 7: (a) Myotube orientation and (b) myotube area at Day 7 for 10% and 2% FBS conditions.

To further characterize the quality of differentiation between conditions, myotube orientation and cross-sectional area were quantified at Day 7 for both 2% and 10% FBS groups

(Figure 7). Mean myotube orientation was approximately 80° for 10% FBS and approximately 90° for 2% FBS, with no statistically significant difference between groups ($p = ns$). Similarly, myotube area was comparable between conditions, with mean values of approximately $1,300 \mu\text{m}^2$ for 10% FBS and approximately $1,450 \mu\text{m}^2$ for 2% FBS, again yielding no significant difference ($p = ns$). While 2% FBS showed a slight trend toward higher mean orientation angle and greater myotube area, the substantial within-group variability in both metrics indicates that serum concentration alone does not produce statistically distinguishable differences in individual myotube morphology or alignment under these conditions. Importantly, the absence of significant differences in orientation and area between groups suggests that when differentiation does occur, the resulting myotubes are of comparable structural quality regardless of whether cells were maintained in 2% or 10% FBS. These findings reinforce the interpretation that 2% FBS achieves equivalent differentiation quality at lower cell density, making it the more controlled and physiologically appropriate condition for subsequent experiments involving macrophage-conditioned media.

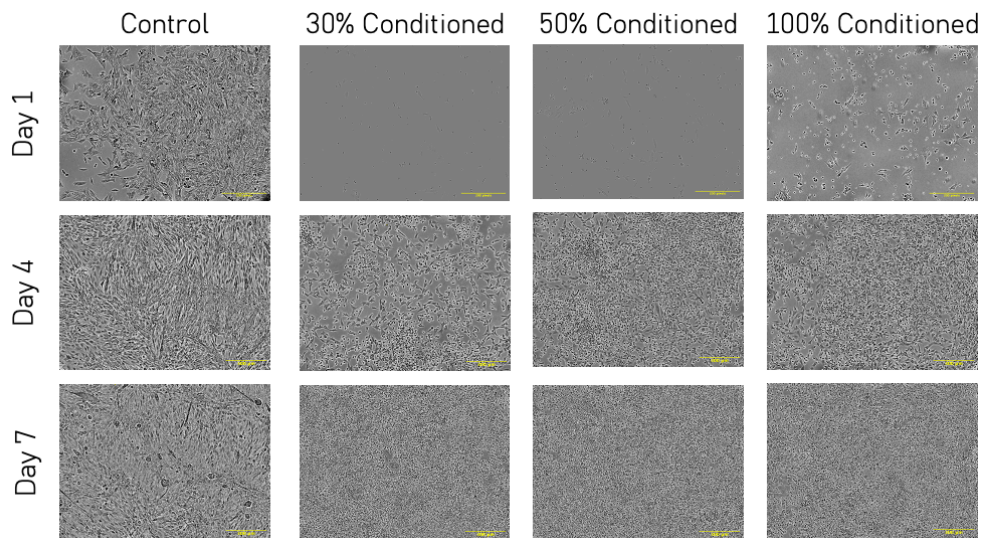


Figure 8: *C2C12 morphology over time (Days 1, 4, 7) under increasing concentration of macrophage-conditioned media (30%, 50%, 100%) compared to control.*

The morphological response of C2C12 myoblasts to increasing concentrations of centrifuged macrophage-conditioned media (MCM) was assessed longitudinally at Days 1, 4, and 7 alongside an unconditioned 2% FBS control (Figure 8). At Day 1, all four conditions — control, 30%, 50%, and 100% MCM — displayed comparable populations of sparse, spindle-shaped myoblasts, indicating that MCM did not immediately alter cell attachment or baseline morphology at the time of seeding. By Day 4, divergence between the control and MCM-treated groups became apparent. The control cultures showed progressive increases in confluence with cells beginning to elongate and cluster, consistent with early myogenic progression. In contrast, the MCM-treated groups exhibited slower proliferative expansion and reduced cell density across all concentrations, with the effect appearing more pronounced at higher MCM fractions. By Day 7, the control cultures had reached substantial confluence with visible elongated cell organization, while MCM-treated wells remained comparatively sparse and lacked the organized, aligned morphology indicative of active myotube formation. Notably, individual cells in the MCM-treated cultures retained a more rounded, less elongated phenotype than the control at this timepoint, suggesting disruption of the normal myogenic morphological program rather than simple cytotoxic cell loss.

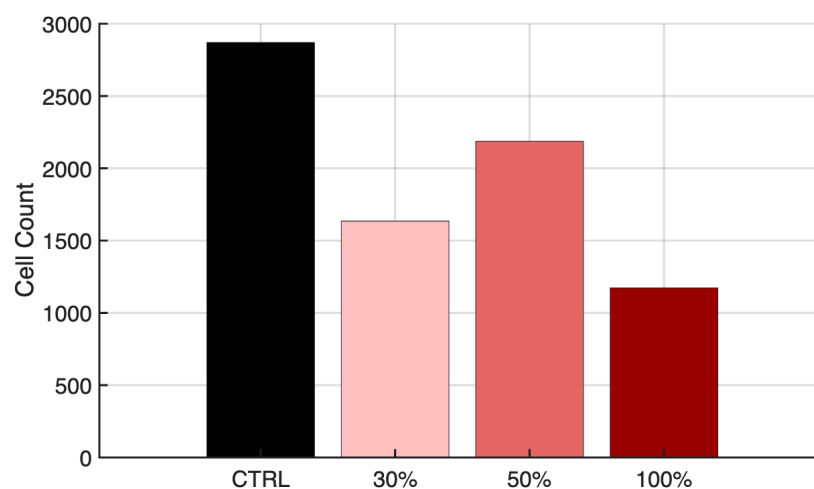


Figure 9: Cell count at Day 7 under increasing concentrations of macrophage-conditioned media.

Cell count quantification at Day 7 revealed a clear concentration-dependent reduction in total cell number across MCM-treated groups relative to the control (Figure 9). The control condition yielded approximately 2,800 cells per field, while the 30%, 50%, and 100% MCM groups contained approximately 1,650, 2,200, and 1,200 cells, respectively. The overall trend indicates that MCM suppresses C2C12 proliferation and/or survival in a dose-dependent manner, with the 100% MCM group showing the greatest reduction. The non-monotonic pattern — where 50% MCM produced a higher cell count than 30% — may reflect variability inherent to conditioned media preparations, including batch-to-batch differences in the concentration of secreted cytokines and growth factors between collections. Regardless of this intermediate deviation, the directional decrease from control to 100% MCM supports the conclusion that soluble macrophage-derived factors exert inhibitory pressure on C2C12 expansion under these culture conditions.

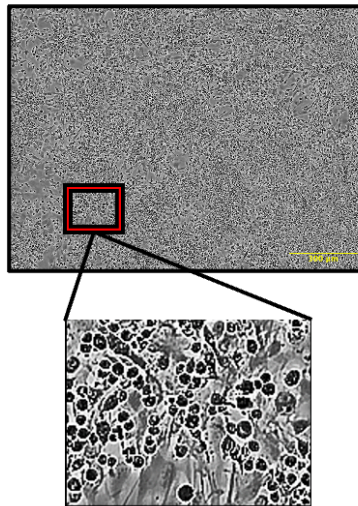


Figure 10: *Representative image of non-myoblast cell populations observed in conditioned media-treated cultures*

A critical methodological observation that informed the design of these MCM experiments was the identification of non-myoblast cell populations within conditioned media-

treated cultures in initial runs (Figure 10). High-magnification imaging revealed morphologically distinct cells — rounded, granular, and adherent — consistent with the appearance of RAW 264.7 macrophages, suggesting that residual macrophages were carried over into C2C12 cultures during media transfer. The presence of these cells would confound interpretation of any MCM effect by introducing direct cell-cell interactions and additional paracrine signals beyond those intended by the conditioned media alone. To eliminate this source of contamination and isolate the contribution of soluble signaling factors, all subsequent MCM preparations were centrifuged prior to use and only the clarified supernatant was applied to C2C12 cultures. This step ensured that the observed effects of MCM on myoblast morphology and cell count in Figures 8 and 9 reflect the influence of secreted soluble factors rather than physical macrophage co-culture.

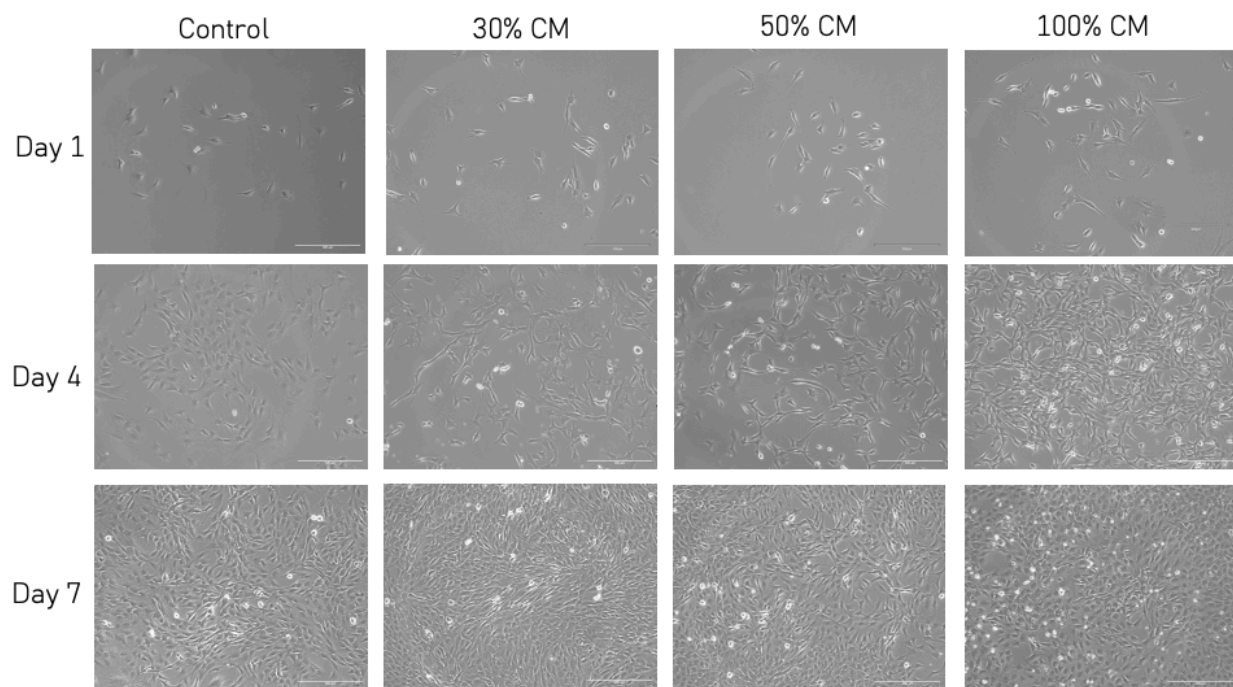


Figure 11: C2C12 morphology over time (Days 1, 4, 7) under increasing concentrations of centrifuged macrophage-conditioned media (30%, 50%, 100%) compared to control

The morphological response of C2C12 myoblasts to centrifuged macrophage-conditioned media was assessed longitudinally at Days 1, 4, and 7 across control, 30%, 50%, and 100% MCM conditions (Figure 11). At Day 1, all groups appeared comparable, with sparse, spindle-shaped myoblasts distributed across the culture surface, confirming equivalent seeding conditions. By Day 4, the control exhibited progressive elongation and increasing confluence consistent with early myogenic progression, while MCM-treated cultures showed visibly reduced cell density and a shift toward more rounded, less organized cell morphologies across all concentrations. By Day 7, these differences were substantially amplified. Control cultures displayed a dense, aligned monolayer with morphological features characteristic of advancing myotube formation. In contrast, MCM-treated wells remained sparser and lacked the directional elongation observed in the control, with the degree of morphological disruption appearing to increase with MCM concentration. Notably, cells in the 100% MCM group showed the most pronounced departure from myogenic morphology, consistent with a concentration-dependent inhibitory effect of macrophage-derived soluble factors on myogenic progression.

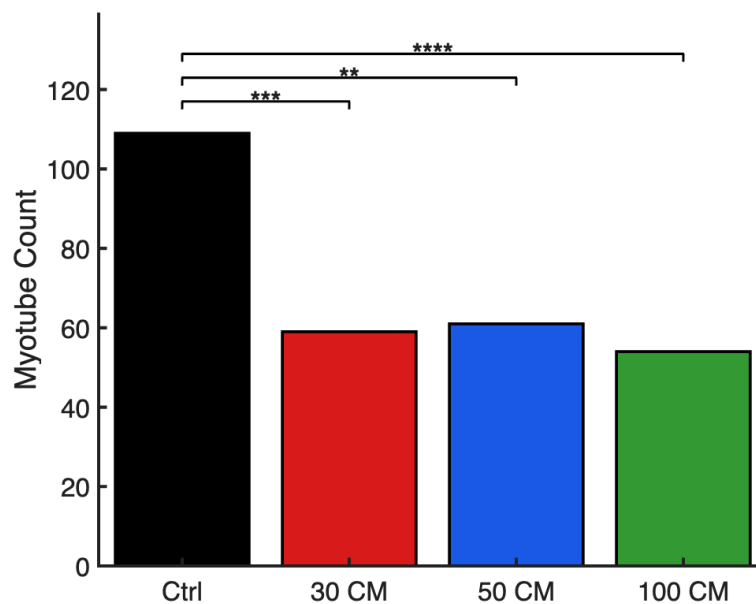


Figure 12: *Quantification of myotube count at Day 7 under increasing concentrations of centrifuged conditioned media.*

Myotube count quantification at Day 7 confirmed and extended the morphological observations, revealing a statistically significant reduction in myotube number across all MCM-treated groups relative to the control (Figure 12). The control condition yielded approximately 110 myotubes per field, while the 30%, 50%, and 100% MCM groups produced approximately 60, 62, and 55 myotubes, respectively — reductions of roughly 45–50% compared to control. Statistical analysis indicated significant differences between the control and all three MCM conditions ($p < 0.01$ for 30% CM, $p < 0.001$ for 50% CM, and $p < 0.0001$ for 100% CM). Notably, the myotube counts among the three MCM groups were similar to one another, suggesting that the inhibitory effect on myotube formation plateaus at relatively low MCM concentrations and does not continue to diminish linearly with increasing dose. This pattern implies that even a partial substitution of differentiation media with MCM is sufficient to substantially impair the myogenic program, and that this effect does not escalate proportionally with MCM fraction beyond the 30% threshold.

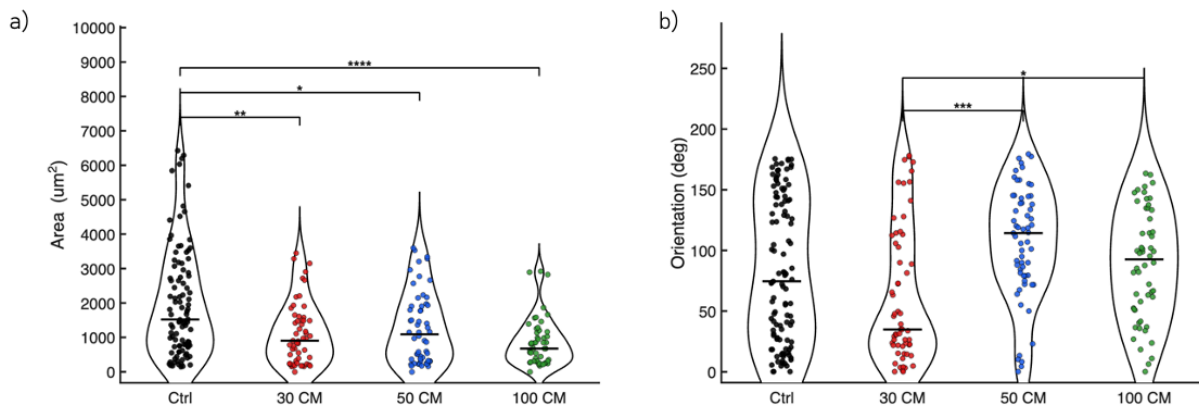


Figure 13: (a) Myotube area and (b) orientation at Day 7 across control and conditioned media groups.

To further characterize the quality and structure of myotubes that did form under MCM conditions, myotube area and orientation were quantified at Day 7 across all groups (Figure 13).

Myotube area was significantly reduced in all MCM-treated conditions compared to control ($p < 0.01$ for 30% CM, $p < 0.05$ for 50% CM, and $p < 0.0001$ for 100% CM), with mean area values decreasing progressively from the control to the 100% MCM group. The 100% MCM group showed the most compact and narrow distribution, indicating that the myotubes that did form were substantially smaller in cross-sectional footprint. Myotube orientation was also significantly altered by MCM treatment, with the 50% CM group displaying a statistically significant shift relative to control ($p < 0.001$) and the 100% CM group showing a significant difference as well ($p < 0.05$). The control group exhibited a relatively broad distribution of orientation angles concentrated in the lower degree range, consistent with partially organized but variable alignment, while the MCM-treated groups displayed wider and more dispersed orientation distributions, reflecting a loss of coordinated directional alignment. Together, the significant reductions in both myotube area and the disruption of orientation uniformity indicate that macrophage-conditioned media not only reduces the number of myotubes formed but also compromises the structural quality and spatial organization of those that do form. This finding suggests that the soluble factors present in unpolarized MCM actively interfere with the cytoskeletal reorganization and cell-cell fusion events that underlie productive myotube maturation in 2D culture.

6.2 Rheologically Tuned Alginate Recapitulates the Mechanical Environment of Early Muscle Regeneration

The findings from the 2D culture experiments established that macrophage-conditioned media disrupts myogenic progression in a conventional planar environment, motivating a transition to a three-dimensional platform where both mechanical and biochemical cues could be independently controlled.

Context	Stiffness	Citation
Healthy Skeletal Muscle	~8-12 kPa	Engler et al., 2006
Regenerating/damaged muscle	~2-5 kPa	Trensz et al., 2015
Hydrogels supporting C2C12 differentiation	~1-3 kPa	Engler et al., 2004

Table 2: *Alginate Tuning*

Skeletal muscle occupies a distinct mechanical niche: healthy mature muscle tissue exhibits a stiffness of approximately 8–12 kPa [16], while the extracellular matrix of regenerating or acutely damaged muscle is considerably softer, falling in the range of approximately 2–5 kPa as the tissue transitions through early inflammatory and repair phases [17]. Critically, hydrogels engineered to recapitulate this softer regenerative range — approximately 1–3 kPa — have been shown to actively support C2C12 myoblast differentiation by providing mechanical cues that promote cytoskeletal organization and myogenic transcriptional activity [18]. As summarized in Table 2, these benchmarks defined the target design window for the alginate hydrogel formulations developed in this work. Alginate was selected as the scaffold material due to its tunable stiffness, cytocompatible ionic crosslinking with calcium chloride, and its established use as a bioink in extrusion-based bioprinting applications. The rheological characterization experiments described below were designed to confirm that the fabricated hydrogels fall within the physiologically relevant stiffness range for modeling early-stage muscle repair

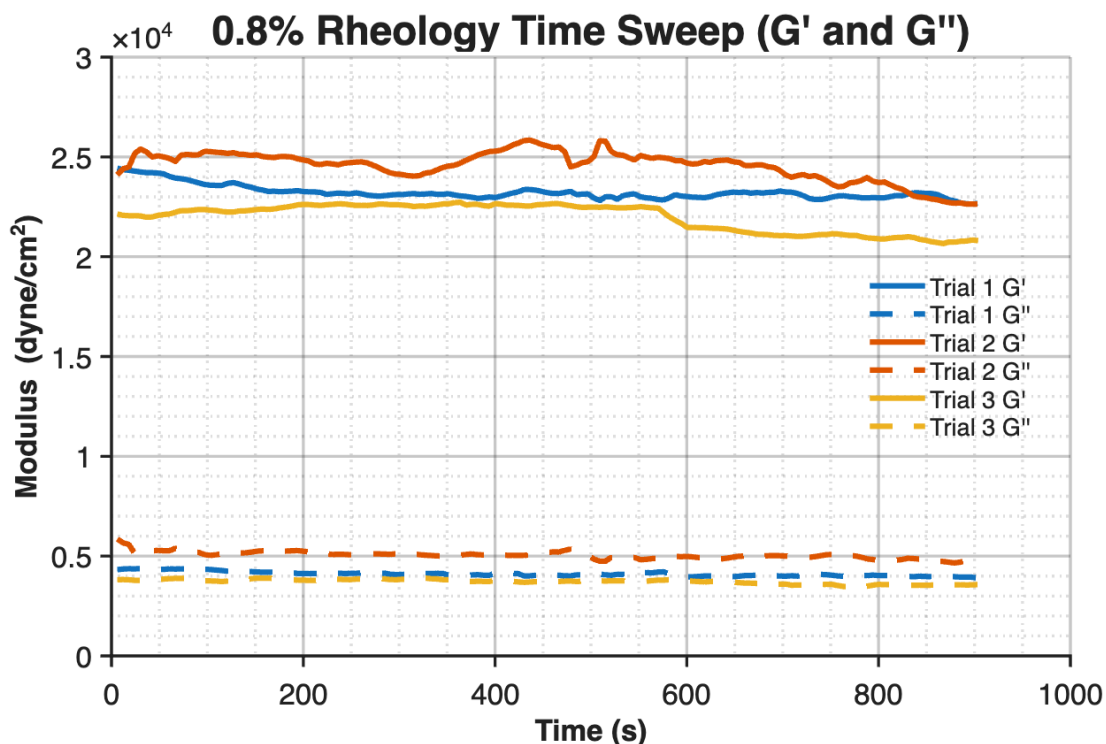


Figure 14: Time sweep rheology of 0.8% alginate showing storage (G') and loss (G'') moduli over time

The viscoelastic properties of 0.8% w/v low-molecular-weight alginate hydrogels crosslinked with calcium chloride were first assessed using oscillatory time sweep rheology conducted across three independent trials (Figure 14). In all trials, the storage modulus (G') remained consistently and substantially greater than the loss modulus (G'') throughout the duration of the 1000-second measurement window, with G' values ranging from approximately 2.2 to 2.5 kPa and G'' values remaining well below 1 kPa across all trials. The persistent dominance of G' over G'' at all timepoints confirms that the hydrogel behaves as a predominantly elastic solid rather than a viscous fluid — a prerequisite for a structurally stable scaffold that can resist deformation under static culture conditions. The stability of both moduli over the full measurement period further indicates that the crosslinked network did not undergo significant structural relaxation, syneresis, or degradation during the timescale of the experiment, supporting the mechanical durability of the formulation.

Trial	Mean Storage G' (kPa)	Standard Deviation
1	2.3223	± 0.040213
2	2.4552	± 0.088684
3	2.1969	± 0.070568

Table 3: Mean storage modulus (G') and standard deviation for each trial

The mean storage modulus values obtained from the time sweep across three independent trials are summarized in Table 3. Trial 1 yielded a mean G' of 2.3223 ± 0.040 kPa, Trial 2 produced 2.4552 ± 0.089 kPa, and Trial 3 yielded 2.1969 ± 0.071 kPa. The narrow standard deviations within each trial confirm that the hydrogel network was mechanically stable throughout the measurement window, with minimal fluctuation in elastic behavior over time. Across the three trials, the inter-trial range of approximately 2.20–2.46 kPa demonstrates strong batch-to-batch reproducibility, indicating that the fabrication protocol — including THINKY centrifugal mixing and ionic crosslinking with calcium chloride — consistently yields hydrogels of predictable mechanical character. This reproducibility is a critical practical requirement for a tissue engineering scaffold, as variability in substrate stiffness between experimental runs would confound interpretation of cell behavior. The collective mean G' of approximately 2.35 kPa places this formulation within the 2–5 kPa stiffness range associated with regenerating and damaged muscle ECM [17] and within the 1–3 kPa range shown to support C2C12 myoblast differentiation on hydrogel substrates [18], confirming that the 0.8% alginate formulation meets the mechanical design criteria established for this platform.

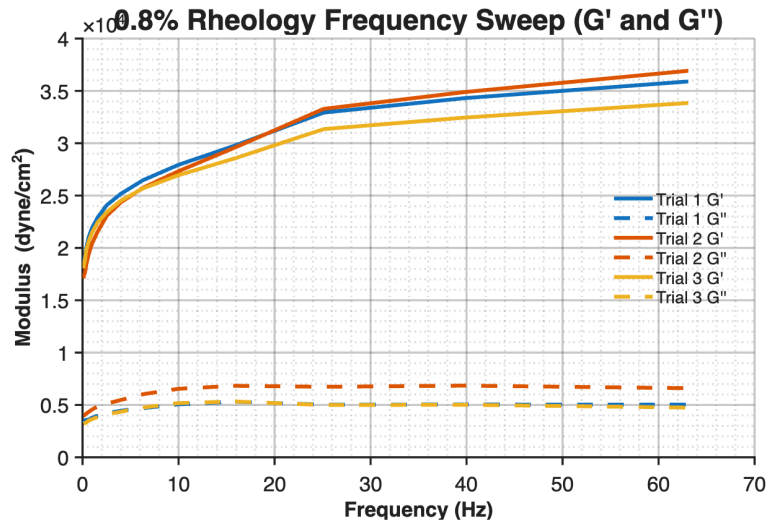


Figure 15: Frequency sweep rheology of 0.8% alginate showing storage (G') and loss (G'') moduli as a function of frequency

To evaluate how the hydrogel mechanical properties respond under dynamic oscillatory loading, frequency sweep rheology was performed across a range of 0.1 to 63 Hz (Figure 15). Across all three trials, G' remained greater than G'' at every tested frequency, reaffirming the predominantly elastic, solid-like character of the crosslinked network established in the time sweep. Both G' and G'' increased gradually with increasing frequency, a response consistent with the viscoelastic behavior of physically crosslinked hydrogel networks, in which the polymer chains have less time to relax and redistribute stress at higher oscillatory rates. Importantly, the G'/G'' ratio remained stable and G' did not drop or plateau abruptly at any frequency within the tested range, indicating that the gel network was not approaching a structural failure or gel-to-sol transition under these conditions. The three G' curves tracked closely with one another across the full frequency range, further reinforcing the inter-trial reproducibility observed in the time sweep data. The modest frequency dependence of G' is also a desirable feature for a cell culture scaffold, as it suggests the hydrogel will maintain its structural integrity across a range of dynamic mechanical perturbations that cells may exert during spreading, migration, or contraction. Together, the time sweep and frequency sweep rheological data provide

complementary evidence that the 0.8% alginate formulation fabricated here is a mechanically reproducible, stable, and predominantly elastic hydrogel whose stiffness is tuned to the physiologically relevant range for modeling the early post-injury muscle microenvironment

6.3 The 3D Alginate Microenvironment Maintains Myoblast Viability Despite Exposure to Conditioned Media

Having established that macrophage-conditioned media significantly disrupts myogenic progression in 2D culture, a critical question arose as to whether the same inhibitory effects would persist when cells are housed within a three-dimensional hydrogel microenvironment. The transition from 2D monolayer to 3D encapsulation represents more than a geometric shift — it fundamentally alters the biophysical context in which cells receive and respond to soluble signals. In a 3D alginate matrix, cells are suspended within a crosslinked network that mechanically supports them on all sides, limits direct surface contact, and may modulate diffusion of macrophage-derived factors to the cell surface in ways that differ substantially from open 2D culture conditions. Alginate itself is a bioinert polysaccharide that lacks integrin-binding motifs, meaning that its protective effects are likely mediated through physical sequestration, altered cytokine diffusion kinetics, and reduced direct membrane exposure rather than active pro-survival signaling [13]. The experiments described in this section were therefore designed to evaluate whether the rheologically tuned 0.8% alginate hydrogel developed in Section 6.2 could support C2C12 myoblast survival over a 7-day culture period in the presence of 50% MCM — the same concentration that produced significant disruption in 2D — using Live/Dead fluorescence imaging as the primary readout.

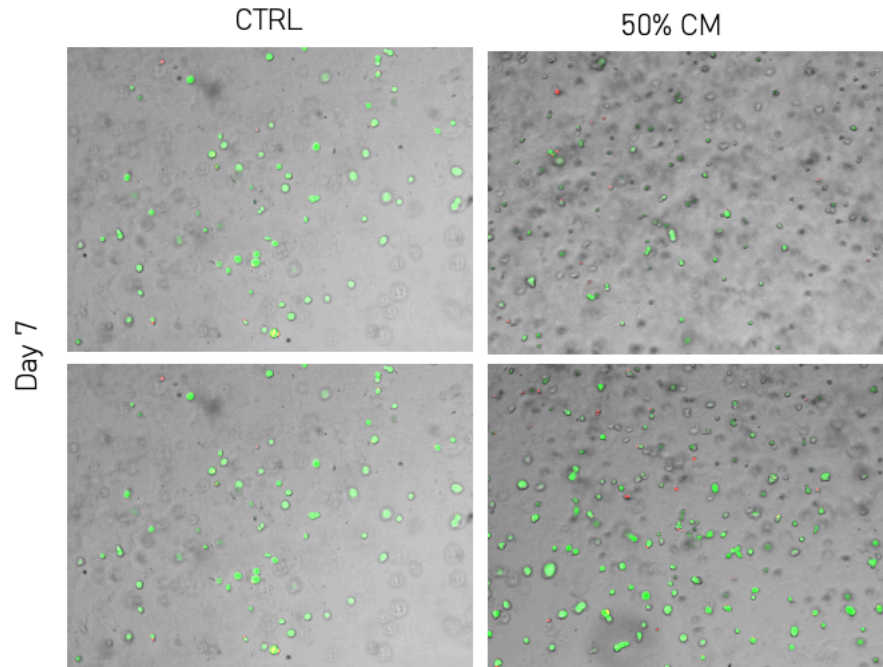


Figure 16: Representative Day 7 live/dead overlay images of C2C12 cells encapsulated in 0.8% alginate hydrogels under control (2% FBS, no CM) and 50% conditioned media conditions.

Cell viability of C2C12 myoblasts encapsulated in 0.8% alginate hydrogels was assessed at Day 7 using Calcein-AM and Ethidium Homodimer-1 fluorescence staining under control (2% FBS, no MCM) and 50% MCM conditions (Figure 16). Representative overlay images from two fields of view per condition reveal robust populations of Calcein-AM-positive (green, live) cells distributed throughout the hydrogel in both groups, with only sparse Ethidium Homodimer-1-positive (red, dead) cells visible across the imaged fields. Importantly, the spatial distribution and density of live cells appeared broadly comparable between the control and 50% MCM conditions at this qualitative level, with no obvious zones of cell death, clustering of dead cells, or degradation of the gel structure apparent in either group. The presence of healthy, membrane-intact cells distributed throughout the three-dimensional gel volume at Day 7 confirms that the alginate encapsulation protocol supports sustained C2C12 survival across the full culture

window and that the hydrogel matrix does not exhibit cytotoxic properties under either condition. Notably, this result stands in stark contrast to the 2D MCM experiments, where 50% MCM produced measurable reductions in cell count and morphological disruption by Day 7, suggesting that the 3D alginate environment confers a degree of cytoprotection against the inhibitory paracrine signals present in MCM.

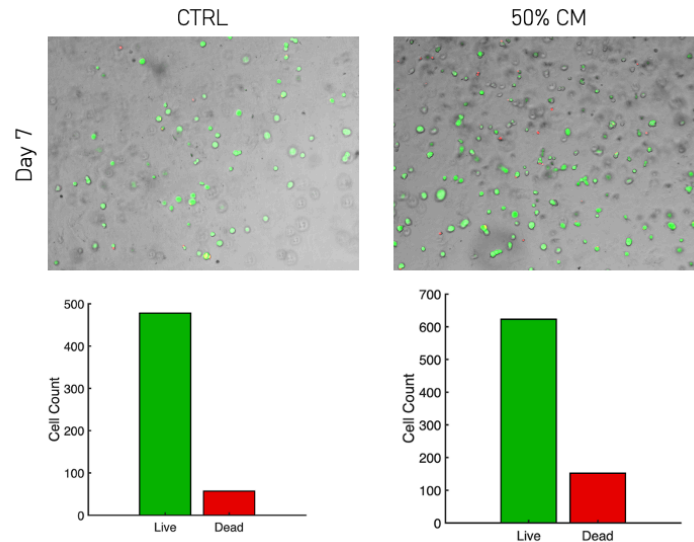


Figure 17: Raw live and dead cell counts for control and 50% CM, generated from a batch of 8 images per condition

Quantitative Live/Dead analysis generated from a batch of 8 images per condition confirmed and extended the qualitative observations from Figure 16 (Figure 17). In the control group, raw cell counts yielded approximately 475 live cells and approximately 75 dead cells per field. In the 50% MCM group, live counts were approximately 625 and dead counts were approximately 150 per field — yielding live fractions of approximately 86% for the control and approximately 81% for the 50% MCM group. Statistical analysis indicated no significant difference ($p = \text{n.s.}$) in live fraction between the two conditions, confirming that the presence of 50% MCM in the gel formulation did not meaningfully compromise cell viability relative to the control. The slightly higher absolute cell counts in the 50% MCM group may reflect variability

in initial seeding density or local gel heterogeneity across imaging fields rather than a true proliferative effect of conditioned media, particularly given that alginate does not support robust cell adhesion or spreading. Taken together, these findings demonstrate that the 0.8% alginate hydrogel platform maintains high C2C12 viability in the presence of macrophage-derived soluble factors, supporting its suitability as a cytocompatible scaffold for future multifactorial studies combining mechanical and immune-derived cues. This cytoprotective behavior is consistent with the established role of 3D hydrogel encapsulation in buffering cells from cytotoxic or disruptive signals in their surrounding media.

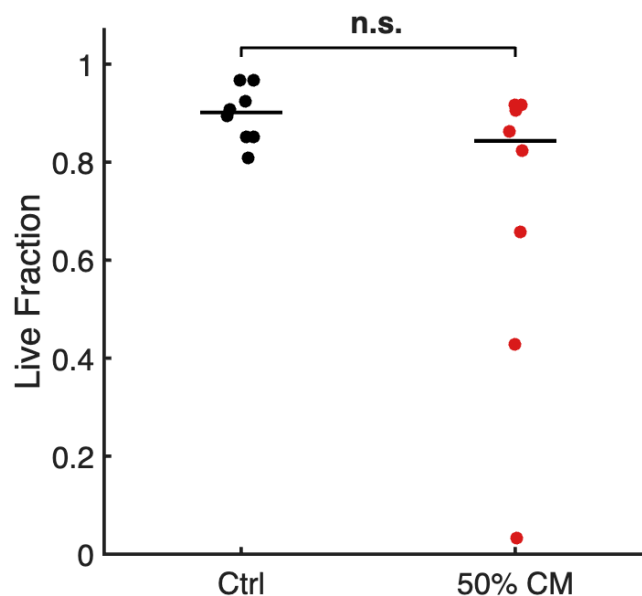


Figure 18: Live fraction analysis indicates no significant difference (*n.s.*) between groups.

To statistically evaluate the viability data across the full image batch, live fraction — defined as the ratio of live cells to total cells counted per image — was calculated for each condition and compared between the control and 50% MCM groups (Figure 18). The control group displayed a tightly clustered distribution of live fraction values centered around approximately 0.90, with minimal spread across the six data points, indicating highly consistent viability across imaged fields. The 50% MCM group showed a mean live fraction of

approximately 0.84, with greater variability among individual data points, including two outlier values at approximately 0.42 and near 0.00 that substantially increased the within-group spread. Despite this variability, statistical comparison between the two groups yielded no significant difference ($p = \text{n.s.}$), confirming that the mean viability of cells encapsulated in MCM-containing hydrogels was not statistically distinguishable from the control at Day 7. The wider distribution observed in the 50% MCM group warrants consideration, however, as the presence of low-viability outlier images suggests that MCM exposure may introduce heterogeneity in the local cellular environment — potentially reflecting uneven distribution of conditioned media factors within the gel matrix or localized regions of reduced nutrient diffusion. Nonetheless, the overall finding that live fraction was maintained at a high mean value in the 50% MCM group, and that this was statistically equivalent to the control, reinforces the conclusion that the 3D alginate scaffold provides a broadly cytoprotective niche capable of supporting C2C12 survival in the presence of macrophage-derived soluble factors.

6.4 Optimized Printing Parameters Yield Structurally Consistent and Scalable Alginate

Constructs

With the 0.8% alginate hydrogel confirmed as mechanically suitable and cytocompatible, the final phase of this project focused on establishing the extrusion-based bioprinting parameters necessary to produce geometrically consistent, scalable constructs from this material. The transition from bench-top cast hydrogels to printed constructs is a critical step toward developing a reproducible tissue model that can be fabricated in defined architectures, scaled across experiments, and ultimately seeded with cells in a spatially controlled manner. In extrusion-based bioprinting, two primary parameters govern the fidelity of deposited filaments: the extrusion speed, which controls the volumetric flow rate of material through the nozzle, and the

The effect of extrusion speed on print fidelity was evaluated at a constant printing speed of 10 mm/s across a range of 1 to 8 mm/s (Figure 19). At the lowest extrusion speed tested (1 mm/s), the PI was approximately 0.67, indicating substantial under-extrusion where insufficient material was deposited relative to the nozzle travel speed, resulting in thin, discontinuous filaments with poor structural integrity. As extrusion speed increased from 1 to 4 mm/s, the PI rose progressively toward 1, reaching an optimal value of approximately 1.01 at 4 mm/s. The corresponding representative print image at this condition shows well-defined, uniformly spaced filaments with consistent line widths across the full printed area, confirming that the PI value of ~ 1 corresponds meaningfully to the best-quality structural outcome. Beyond 4 mm/s, the PI dropped slightly to approximately 0.93–0.94 in the 5–7 mm/s range before rising sharply to approximately 1.18 at 8 mm/s, indicating a transition into over-extrusion where excess material was being deposited faster than the print head could lay it down cleanly. These results identify 4 mm/s as the optimal extrusion speed at a printing speed of 10 mm/s, yielding the best balance between material deposition and nozzle travel that produces structurally faithful filaments.

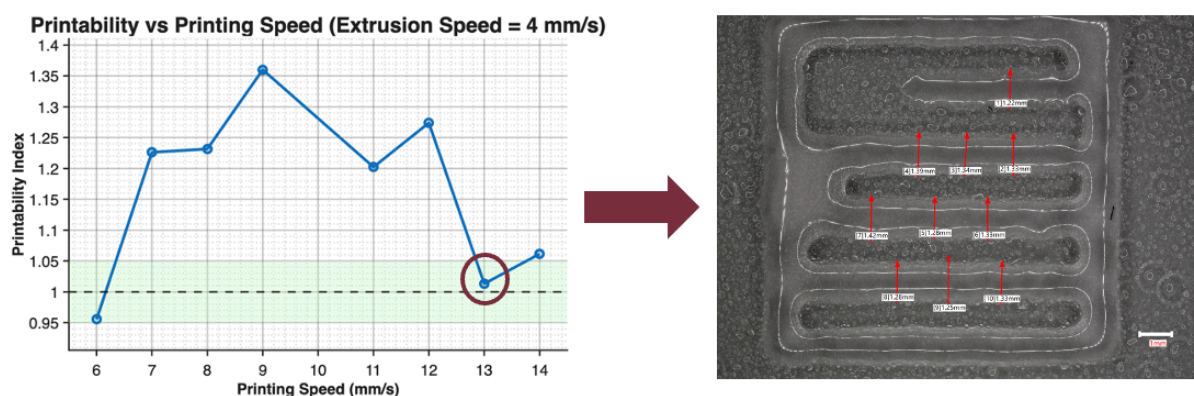


Figure 20 : *Printability index as a function of printing speed at a constant extrusion speed of 4 mm/s.*

With the extrusion speed fixed at the identified optimum of 4 mm/s, the effect of printing speed was evaluated across a range of 6 to 14 mm/s (Figure 20). At lower printing speeds (6–12 mm/s), the PI values were consistently elevated above 1, ranging from approximately 0.95 at 6 mm/s to values as high as approximately 1.36 at 9 mm/s, with considerable variability across the tested range. These elevated PI values at slower printing speeds reflect over-extrusion conditions in which the print head moves too slowly relative to the volumetric output of the nozzle, causing material to accumulate and spread beyond the target filament width. As printing speed increased toward 13 mm/s, the PI dropped sharply to approximately 1.01, closely approaching the target value of 1. The corresponding print image at 13 mm/s shows uniformly deposited filaments with well-defined edges and consistent inter-filament spacing across all four rows of the printed construct, visually corroborating the quantitative PI result. At 14 mm/s, the PI rose slightly to approximately 1.05, suggesting the onset of mild under-extrusion as the nozzle began to outpace material deposition. Together, these results identify 13 mm/s as the optimal printing speed when operating at an extrusion speed of 4 mm/s. Taken in combination, the parameter optimization experiments establish 4 mm/s extrusion speed and 13 mm/s printing speed as the conditions that consistently yield alginate filaments with the highest structural fidelity, providing a reproducible and scalable bioprinting protocol for this hydrogel formulation.

7. CONCLUSION

This thesis aimed to develop and characterize a rheologically tuned alginate hydrogel platform that integrates macrophage-conditioned media to recapitulate key features of the early skeletal muscle regeneration niche *in vitro*. The work was carried out in three sequential experimental phases — 2D culture optimization, 3D hydrogel development and cell

encapsulation, and bioprinting parameter optimization — each building directly on the findings of the previous. Taken together, the results of this study establish a reproducible, biofabrication-compatible platform that simultaneously addresses both the mechanical and biochemical dimensions of early-stage muscle repair modeling.

The first phase of this work established that 2% FBS is the optimal serum concentration for supporting myogenic differentiation of C2C12 myoblasts in 2D culture. At this concentration, cells reliably progressed through the early stages of the myogenic program, exhibiting characteristic elongation, alignment, and the onset of myotube organization by Day 7, while maintaining adequate viability throughout the culture window. Complete serum withdrawal at 0% FBS was insufficient to sustain productive differentiation due to its cytotoxic effects on cell density, while 10% FBS maintained cells in a predominantly proliferative state that was not conducive to terminal differentiation. These findings established 2% FBS as the baseline differentiation condition for all subsequent experiments.

The introduction of macrophage-conditioned media (MCM) in the second phase of the 2D experiments revealed that soluble macrophage-derived factors significantly disrupt myogenic progression. Across all MCM concentrations tested — 30%, 50%, and 100% — myotube count was significantly reduced relative to the unconditioned control, and myotubes that did form were smaller in area and less directionally aligned. Critically, these effects were observed after the conditioned media was centrifuged to remove residual macrophage cells, confirming that the disruption was attributable to soluble paracrine signaling factors rather than direct macrophage contact. This finding has important implications: it suggests that the MCM used in these experiments, derived from unpolarized RAW 264.7 macrophages, did not recapitulate the pro-

differentiation signaling associated with M2 macrophage polarization that is characteristic of the later regenerative phase in vivo. Rather than reflecting a simple cytotoxic effect, the suppression of myogenesis observed here likely reflects the absence of appropriate polarization cues and underscores the importance of macrophage phenotype in future immune-integrated muscle models.

Moving into the three-dimensional platform, 0.8% low-molecular-weight alginate hydrogels crosslinked with calcium chloride were fabricated and characterized by oscillatory rheology. Time sweep and frequency sweep analyses across three independent trials confirmed that the hydrogel exhibits predominantly elastic, solid-like behavior with a mean storage modulus of approximately 2.2–2.5 kPa — a range that falls within the reported stiffness of regenerating muscle ECM and that has been shown to support C2C12 differentiation on hydrogel substrates. The reproducibility of these values across trials demonstrated that the THINKY centrifugal mixing and ionic crosslinking protocol consistently produces mechanically defined hydrogels suitable for use as a cell culture scaffold. When C2C12 myoblasts were encapsulated in these hydrogels and cultured for seven days in the presence of 50% MCM, Live/Dead staining and live fraction analysis revealed high cell viability in both the control and MCM-treated groups, with no statistically significant difference between conditions. This result stands in clear contrast to the cytotoxic and differentiation-suppressive effects of 50% MCM observed in 2D, suggesting that the three-dimensional alginate microenvironment confers a degree of cytoprotection against disruptive immune-derived signals, supporting its translational relevance as a tissue model substrate.

Finally, extrusion-based bioprinting parameters were systematically optimized using a Printability Index defined as the ratio of the measured deposited filament width to the optimal width — where the optimal width was determined empirically by identifying the prints with the best structural fidelity and calculating their average filament dimensions. Evaluation across a range of extrusion and printing speeds identified 4 mm/s extrusion speed and 13 mm/s printing speed as the parameter combination that most consistently produced filaments with a Printability Index approaching 1, corresponding to uniform, well-defined constructs with consistent inter-filament spacing. These optimized parameters establish a reproducible bioprinting protocol for the 0.8% alginate formulation and confirm its suitability as a printable bioink for scalable construct fabrication.

Several limitations of the current platform must be acknowledged. First, C2C12 myoblasts are inherently adherent cells, and alginate's bioinert surface provides few structural anchoring points, which is likely limiting cell spreading, cytoskeletal organization, and myotube fusion in the 3D environment. Second, conditioned media composition can vary between macrophage culture flasks due to differences in passage number, cell density at collection, and culture history, introducing batch-to-batch inconsistency in the immune signals delivered across experiments. Third, skeletal muscle stiffness changes dynamically throughout the regeneration process as cells mature and myotubes develop — a static hydrogel cannot replicate this evolving mechanical landscape, which is why this model is intentionally scoped to the early stages of repair. To address these limitations in future work, incorporation of gelatin or collagen into the alginate matrix is proposed to introduce cell-adhesive motifs that would better support myoblast attachment and fusion in 3D. Standardizing MCM collection to a fixed timepoint following macrophage thawing or passaging every time would reduce variability in conditioned media

composition across experiments. Additionally, tuning the alginate percentage and crosslinking duration toward a more permissive formulation could create a less restrictive mechanical environment to better support myogenic progression within the gel.

In summary, this work successfully established a tunable alginate-based in vitro model of early-stage muscle repair, demonstrated that 2% FBS is the optimal baseline for myogenic differentiation, showed that unpolarized macrophage-conditioned media disrupts both cell behavior and myotube formation in 2D, confirmed that the 3D alginate scaffold maintains cell viability even in the presence of immune-derived signals, and identified reproducible bioprinting parameters for scalable construct fabrication. Together, these findings advance the development of a biofabrication-ready platform that begins to integrate the mechanical and biochemical complexity of the early regenerative muscle niche, and highlight the need for future models that combine precisely tuned substrate mechanics with well-defined, polarization-controlled immune cues to more fully capture the biology of skeletal muscle repair.

8. FUTURE WORK

The findings of this thesis establish a functional foundation for a mechanically and biochemically tunable in vitro muscle repair model, but several important experimental directions remain to be pursued to fully realize the platform's potential.

The most immediate and consequential next step is the polarization of RAW 264.7 macrophages into defined M1 and M2 phenotypes prior to conditioned media collection. In the native regenerative sequence, infiltrating macrophages transition from a pro-inflammatory M1 state — characterized by secretion of cytokines such as IFN- γ , TNF- α , and IL-6 — to an anti-

inflammatory M2 state driven by IL-4 and IL-13 signaling, which promotes myoblast differentiation, myotube fusion, and tissue remodeling. The conditioned media used in the current study was derived from unpolarized macrophages, meaning it contained a mixed and uncontrolled cytokine milieu that did not faithfully represent either phase of the regenerative immune response. By generating stage-specific M1- and M2-conditioned media and applying each independently to C2C12 cultures, future experiments will be able to directly compare how pro-inflammatory versus anti-inflammatory paracrine environments differentially regulate myoblast differentiation, fusion index, and myotube maturation. This would provide mechanistic clarity on which immune signals are responsible for the disruption of myogenesis observed in the current study and whether M2-conditioned media is sufficient to rescue or enhance differentiation in both 2D and 3D systems.

In parallel, Myosin Heavy Chain (MHC) immunostaining should be implemented in both the 2D and 3D experimental contexts to provide a direct molecular readout of terminal differentiation beyond the morphological and phase-contrast-based assessments used in this work. MHC expression, detected via the MF20 antibody with DAPI nuclear counterstaining, would enable quantification of fusion index — the fraction of nuclei contained within MHC-positive multinucleated cells — across serum conditions, MCM concentrations, and polarization states. This metric is the gold standard for assessing functional myogenic differentiation in vitro and would substantially strengthen the conclusions drawn from the current morphological data. Applying MHC staining within the 3D alginate constructs would additionally confirm whether encapsulated myoblasts are capable of progressing through terminal differentiation within the hydrogel matrix, which has not yet been directly demonstrated.

Ultimately, the overarching goal that this entire platform has been building toward is the quantification of myotube formation within the three-dimensional hydrogel itself. Demonstrating that C2C12 cells can not only survive but actively fuse and form mature myotubes within a rheologically tuned, MCM-supplemented alginate construct would validate the platform's core premise — that a combined mechanical and immune microenvironment can recapitulate key features of early skeletal muscle repair in vitro. Achieving this milestone would position the system as a functional tissue model with genuine translational relevance for studying immune-muscle communication, screening candidate regenerative therapies, and reducing reliance on in vivo animal models for early-stage repair research.

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